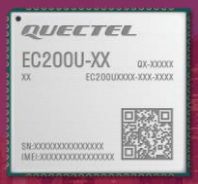
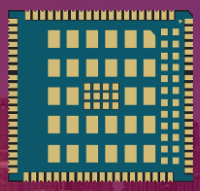
Quectel EC200U Series

IoT/M2M-optimized

LTE Cat 1 Modules





EC200U-CN-AA

Release Notes

LTE Standard Module Series

Rev. EC200U-CN-AA_Firmware_Release_Notes_V0313

Date: 2024-12-10

Our aim is to provide customers with timely and comprehensive service. For any assistance, please contact our company headquarters:

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1. Release Content

This document provides the Release Notes for EC200U-CN-AA. The current release includes the firmware package.

Package	Version
Firmware	EC200UCNAAR03A13M08

2. Matters Needing Attention

SN	Item
[1]	In the previous version, when PWRKEY has been connected to the ground, the module cannot be turned on again after it is turned off. Only after a complete power-down can the module be turned on. It is supported that the module can be turned on after turning it off by **AT+QPOWD=2**, but the current leakage becomes higher.
[2]	The PDP context activated with **AT+CGACT**, **AT+QIACT** or data call API requires de-activation to configure a new APN.
[3]	Provide power supply to VBUS before turning on the module, and then turn VBAT on. It is recommended to wait for at least 2 seconds after VBAT pulls up and then pull down PWRKEY to turn on the module, and PWRKEY should be pulled down for at least 2 second.
[4]	Due to the small UFS space, when the delta firmware package is upgraded directly in UFS, if the package is too large, you can reduce its size by modifying the profiles such as by ignoring the preset files. If the delta firmware package is still too large after modification, please divide the package into several smaller ones for FOTA upgrade.
[5]	Hardware flow control of the serial port does not support the MCU to pull RTS at any time, but only to pull RTS during the stop bit, otherwise the module will receive junk data.
[6]	When the USB is suspended, the USB reset signal of the host cannot be identified.
[7]	To extend the service life of flash, it is recommended that the total number of operations related to powering on/off, CFUN switching, SIM card hot plugging, or dual-SIM switching should not exceed 30 times per day.
[8]	"Execute **AT+QMTDISC** to disconnect from the MQTT server, the server responds and actively shuts down, and the module reports the shutdown status", this optimization was added in EC200UCNAAR03A07M08 or above.
[9]	Because the software code varies between different versions released on across-baseline, versions released on R05 baseline and versions released on R02/R03 baseline cannot

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	perform DFOTA each other. In principle, it is not recommended to upgrade versions released on cross-baseline to each other. If you have this requirement, please be prepared for local upgrade and conduct relevant tests in advance.
[10]	Preset files for functions such as GNSS, BT and TTS in the file system cannot be deleted at will.
[11]	The module turns on the remap function by default. This function is suitable for most operators' LTE network scenarios.
[12]	Imported the GNSS firmware ROM2 from the version EC200UCNAAR03A09M08 or above. And the module with OC number of EC200UCNAP-N05-SGNSA and EC200UCNAP-N05-MN0CA requires to use the version EC200UCNAAR03A09M08 or above.
[13]	If the embedded protocol stack and the external protocol stack use the same CID, you need to execute **AT+QCFG="nat/cid"** to enable NAT. For details, please contact Quectel Technical Support.
[14]	During the FOTA upgrade process, it is necessary to ensure the stable power supply of the module. If the power is disconnected during the upgrade, there is a small probability that the flash will be damaged or other abnormalities may occur.

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3. Release History

3.1. Firmware Release History

Firmware Version	Description
EC200UCNAAR03A13M08	Mass production.
EC200UCNAAR03A12M08	Mass production.
EC200UCNAAR03A11M08	Mass production.
EC200UCNAAR03A09M08	Mass production.
EC200UCNAAR03A08M08	Mass production.
EC200UCNAAR03A07M08	Mass production.
EC200UCNAAR03A06M08	Mass production.
EC200UCNAAR03A05M08	Mass production.
EC200UCNAAR03A04M08	Mass production.
EC200UCNAAR03A03M08	Mass production.
EC200UCNAAR03A02M08	Mass production.
EC200UCNAAR03A01M08	Mass production.

3.2. New Features

EC200UCNAAR03A13M08	
Item	**Brief Description**
SIMCARD	Added **AT+QIMSCFG="isim"** command to turn on/off ISIM applications.
EC200UCNAAR03A12M08	
Item	**Brief Description**
HTTP	Added a patch request method for HTTP.

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SMS	Added **AT+QREFUSECS** command for rejecting incoming calls/SMS.
GENERAL	Added module as USB HID slave function.
GENERAL	XLAT was compatible with network configuration IPV4, IPV6 types. If IPV4 was all zeros, XLAT was also allowed to be enabled.
EC200UCNAAR03A11M08	
Item	**Brief Description**
AUDIO	Added **AT+QAUDCFG="iic"** to configure the iic channel of the external codec.
PPP	Supported waking up the host through RI pin after receiving the data from the network at PPP dial-up in the sleep state.
TCP SSL	Adds ALPN configuration for SSL.
TCP/UDP	Added **AT+QCFG="tcp/mss"** to support modifying TCP MSS length.
GENERAL	Added USB MSC mode to the virtual disk function, and the implementation mode is consistent with the card reader.
EC200UCNAAR03A09M08	
Item	**Brief Description**
HTTP	Added patch request method for HTTP.
GENERAL	Supported serial port baud rate of 10400.
EC200UCNAAR03A08M08	
Item	**Brief Description**
GNSS	Added **AT+QGPSCFG="urc"** command to enable or disable URC report related to firmware/APflash/AGPS/mode for GNSS.
HTTP	Added HTTP multi-form function.
GENERAL	Added shielding USB remote wake-up function.
EC200UCNAAR03A07M08	
Item	**Brief Description**
AUDIO	Supported using **AT+QAUDPLAY** to play audio files when calls incoming. After the call was answered or hung up, the playing stopped automatically.
NETWORK	Supported obtaining long string name and short string name of TELCEL operators.
NETWORK	Supported configuring an IP address and a fixed port.
EC200UCNAAR03A06M08	

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Item	Brief Description
/	/
EC200UCNAAR03A05M08	
Item	**Brief Description**
DSSS	Added **AT+QDSTYPE** to switch to DSSS and DSDS.
NETWORK	Added **AT+QCFG="creg/mode"** to control **AT+CREG** to get CS registration status.
NETWORK	Added **AT+QCEERCATCFG** to configure response format of **AT+CEER** in 2G/4G.
SIMCARD	Added **AT+QSPN** to get the operator name in unregistered network status.
EC200UCNAAR03A04M08	
Item	**Brief Description**
AUDIO	Added audio parameter calibration commands below: **AT+QAUDCFG="caccp"** **AT+QAUDCFG="cavqe"** **AT+QAUDCFG="capre"** **AT+QAUDCFG="capost"** **AT+QAUDCFG="cawtf"**
GENERAL	Supported 1200 bps UART baud rate.
SMS	Supported the white list for Voice Call and SMS.
TCP/UDP	Added the resource application release interface *ql_malloc()* or *ql_free()*.
EC200UCNAAR03A03M08	
Item	**Brief Description**
NETWORK	Added AT command **AT+QCFG="cops_no_mode_change"** to enable or disable whether to switch the network mode after **AT+COPS=1** execution.
GENERAL	Added AT command **AT+QCFG="urc/ri/ring","auto"** to support pulling the RI pin to low when enabling RING, and pulling the RI pin to high when exiting from RING.
EC200UCNAAR03A02M08	
Item	**Brief Description**
NETWORK	Relevant technical controls have been carried out to restrict normal network registration in regions such as RUS and IRN, thus ensuring that the module can be used only for civilian applications
GENERAL	Added AT command **AT+QCFG="bip/status"** and URC +QBIPSTAT: x to notify customers of BIP status.

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GENERAL	Added AT command **AT+QUSBCFG=3,1** to enumerate USB CDC ACM port (ttyACM0 in Linux).

EC200UCNAAR03A01M08	
Item	**Brief Description**
/	/

3.3. Improved Features

EC200UCNAAR03A13M08	
Item	**Brief Description**
BT	Fixed the problem of incomplete URC reporting when there are many characteristics scanned on the client side.
DTMF	Added the level parameter for **AT+QDTMFDETSETEX** command to configure the detection power.
NETWORK	Fixed the problem of APN set by **AT+CGDCONT** displaying exception after the module rebooted.
NETWORK	Solved the problem that **AT+CGPADDR** could not get the address after **AT+QNETDEVCTL** dialing in some scenarios.
NETWORK	Optimized the **AT+CEER** command.
NETWORK	Solved the problem that the network registration was rejected by some operators.
NETWORK	Solved the problem that SS business of some SIM cards did not use IMS.
NETWORK	Optimized remap logic.
SIMCARD	Optimized the power-down timing of SIM card and increased the redundancy time to operate SIM card.
TCP/UDP	Optimized the problem that quickly executing **AT+QICLOSE** after receiving the Socket service disconnection URC may led to UART being stuck a very small probability.
GENERAL	Removed the firewall function.
GENERAL	Optimized hardware flow control to solve the problem of probabilistic abnormal data in RX direction when hardware flow control was turned on at high baud rates.
GENERAL	Supported accessing the lwip protocol stack of the module through ECM/RNDIS in no SIM card scenarios.
GENERAL	Solved the problem that the mask set by **AT+QCFG="nat/subnet"** did not take effect.

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GENERAL	Optimized the **AT+COPS** command.
GENERAL	Fixed the problem that the IPv6 prefix and Bear ID did not correspond in multiple PDP scenarios for XLAT.
GENERAL	Fixed the problem that an error may occur in continuous configuration of **AT+QPWBACKOFF** command.

EC200UCNAAR03A12M08	
Item	**Brief Description**
BT	Fixed the wrong parameter range for BT-related AT test commands.
BT	Fixed the problem related to BT A2DP&AVRCP audio playback.
BT	Fixed the problem of abnormal BT SPP transparent transmission.
CMUX	Fixed the problem that non-CMUX frame data was abnormally passed after enabling CMUX port.
DFOTA	Optimized the problem that HTTP username and password could not be input in fota_http.
DTMF	Added the level parameter for **AT+QDTMFDETSETEX** command to configure the detection power.
DTMF	Modified the range of time supported by **AT+QWDTMF** from (500-1000) to (10-1000).
FILE	Optimized modem partition mount and SFFS file system.
GNSS	Modified the error code reported after **AT+QGPSLOC=0/1/2** from 50 to 505 when the GNSS was not turned on.
HTTP	Fixed the problem that after the output response header was turned on, the Content-Length returned by the server was 0, there was no Content-Length reported by URC, and **AT+QHTTPREAD** could not read the header information.
NETWORK	Solved the problem that **AT+CGPADDR** could not get the address after **AT+QNETDEVCTL** dialing in some scenarios.
NETWORK	Solved the problem that SS business of some SIM cards did not use IMS.
NETWORK	Solved the problem that the network registration was rejected by some operators.
NETWORK	Optimized remap logic.
NETWORK	Solved the problem that the 2G roaming SIM card could not be registered normally.
PPP	Optimized PPP dial-up function.
SIMCARD	Added SIM card file write protection interface, optimized the power-down timing of the SIM card and increased the redundancy time of SIM card operation.
SIMCARD	Solved the problem of not recognizing 2G SIM cards.
SIMCARD	Solved the problem that in the locked pin status, some SIM cards status obtained by **AT+CPIN?** was incorrect.

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TCP/UDP	Solved the problem that the UDP connection may disconnect abnormally in low probability when sending and receiving a large amount of data quickly.
TCP/UDP	Fixed the problem that after entering BYPASS mode by **AT+QISEND**, and pulling up and down DTR, executing **AT+QISEND** may not exit from BYPASS mode.
TCP/UDP	Optimized URC information reported by the AT+QPING command.
GENERAL	Optimized the **AT+COPS** command.
GENERAL	Fixed an issue where the default bearer became empty after performing AT&F.
GENERAL	Allowed **AT+QWSCFG="write/buffersize"** to set the maximum data length sent by executing **AT+QWSWRITE** once, with the transmitted data being a complete frame.
GENERAL	Fixed the problem that an error may occur in continuous configuration of **AT+QPWBACKOFF** command.
GENERAL	Fixed the problem that DNS could not be resolved correctly.

EC200UCNAAR03A11M08	
Item	**Brief Description**
AUDIO	Solved the problem that audio files could not be played normally.
AUDIO	Fixed the problem of no audio output after repeated configuration as external Codec.
CMUX	Fixed the blocking problem caused by CMUX receiving abnormal data frames.
DTMF	Fixed the problem that DTMF/TONE sound could not be interrupted in real time.
GNSS	Fixed the problem of no NMEA statements outputs if GNSS was restarted in the case of deleting GNSS firmware.
HTTP	Optimized the data download range for the **AT+QHTTPGETEX** command to request a redirect address for the first time.
MQTT	Modified the maximum string length of parameters <clientID>, <username>, <password> in **AT+QMTCONN** command to 256.
PPP	Fixed the problem that PPP dialing by Host failed in some scenarios.
NETWORK	Fixed the problem that the arp packet may fail to be sent in ECM/RNDIS dialing mode.
GENERAL	Optimized the problem of receiving redundant data when UART used DMA mode.
GENERAL	Supported LWIP to distinguish the MTU parameter of IPV4 and IPV6.
GENERAL	Optimized the process for *ql_mkdir()* function implementation.
GENERAL	Fixed a DNS request failure caused by an inconsistency between the domain name address returned by the DNS server and the local domain name address.
GENERAL	Optimized BIP function under IPv6.

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EC200UCNAAR03A09M08	
Item	**Brief Description**
AUDIO	Fixed the problem that if the audio file/audio stream/TTS/DTMF/TONE was interrupted during playing, the module may be abnormal.
BT	Solved the problem of high current consumption when the module entered sleep in Bluetooth mode and used USB aplog to output.
DTMF	Modified the time range supported by **AT+QWDTMF** from (500-1000) to (10-1000).
DTMF	Fixed DTMF */# pronunciation problem.
ECM	Fixed the problem that IPV6 address was obtained abnormally during Android ECS/RNDIS dialing.
FILE	Modified commands such as **AT+QFDEL="UFS:*"** and **AT+QFDEL="*"** to not to delete system preset files such as TTS resource files, GNSS firmware files, and Bluetooth firmware files when you deleted all files in the file system.
FILE	Fixed the problem of file operation on the drive letter when it was not mounted.
FTP	Optimized FTP GET functionality.
GNSS	Optimized GNSS to increase adaptive methods.
GNSS	Modified the error code returned by **AT+QGPSLOC** from 50 to 505 when GNSS function was not turned on.
TCP/UDP	Optimized URC information reported by **AT+QPING**.
GENERAL	Optimized BIP function under IPv6.
EC200UCNAAR03A08M08	
Item	**Brief Description**
AUDIO	Fixed the problem that pcm_clk pcm_sync was pulled low if there was no audio playing when external codec was used.
BT	Fixed the problem that there was no sound in playing when Bluetooth A2DP function was turned on again.
ECM	Fixed the problem that the DNS server could not be accessed normally when USBnet was used and the DNS address of Host was set as the gateway address.
FILE	Fixed the problem that during re-mounting, a wild pointer problem would occur when the FAT file system used spaces that was used in the last mount failure.
GNSS	Optimized GNSS positioning performance.
GNSS	Modified the GNSS AT command response format from **+CMS: xxx** to **+CME: xxx**.
HTTP	Fixed the problem of failed redirection to different resources within the same HTTP server address.
MQTT	Fixed the problem that the application layer received duplicate ID packets if MQTT

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	subscribed to topics with QOS=2.
NETWORK	Fixed the problem that **AT+QOPS** returned the wrong operator name in some scenarios.
SMS	Optimized the problem of inaccurate "smsfull" indication of **AT+CIND** command.
TCP SSL	Fixed the problem that the module failed to connect to the NLB-UDP server.
TCP/UDP	Fixed the problem that AT port could not work normally under poor network environment when **AT+QIOPEN** was in transparent transmission mode.

EC200UCNAAR03A07M08

Item	Brief Description
DFOTA	Fixed a failure to download FOTA files from an IPV6 server by **AT+QFOTADL**.
ECM	Fixed the problem that when USBnet was used and the DNS address of Host was set to the gateway address, the DNS server could not be accessed normally.
FTP	Modified the error code reported when the file was re-opened from 617 to 611.
FTP	Fixed the problem that the module could not work normally when files were downloaded to USB port through AT+QFTPGET.
GENERAL	Fixed the problem that some hosts may not recognize USB after rebooted.
GENERAL	Modified the no-frok parameter in INVITE signaling to fork when VoLTE calls were made.
Jamming Detection	Fixed the problem of abnormal configurations on Jamming Detection.
MQTT	Fixed the problem that the module may not work normally in a possibility during MQTT retransmission.
MQTT	Fixed the problem of data mess occurred when massive MQTT data was sent at the same time.
NTP	Fixed the problem related to NTP thread security.
PING	Fixed the problem that after the socket connection failure or the Ping packet sending failure, AT+QPING execution failed.
SIMCARD	Fixed the problem that data transmitting would be abnormal if SPI (with DMA mode), SD(EMMC), CAMERA and SIM card were used at the same time.

EC200UCNAAR03A06M08

Item	Brief Description
DFOTA	Solved the problem that network registration was affected by repeatedly encrypted calibration parameter after FOTA upgrade between specific versions.
NETWORK	Solved the problem that some of SIM cards of Orange operator was slow to register to the network.

EC200UCNAAR03A05M08

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Item	Brief Description
NETWORK	Fixed the problem that the domain name could not be pinged successfully during host dial-up after NAT and DMZ are enabled.
NETWORK	Fixed the problem of the information returned by **AT+QNWINFO** in special scenarios such as TAU or PSM.
EC200UCNAAR03A04M08	
Item	**Brief Description**
DFOTA	Optimized the filtering judgment mechanism for abnormal events in HTTP FOTA under poor network environment.
DTMF	Optimized DTMF sensitivity during the call under 2G network.
GENERAL	Fixed the problem that data lost when other UARTs (expect for main UART) entered sleep during data sending.
FTP	Solved the problem that some servers failed to obtain files.
FTP	Solved the problem that under dual SIM mode, switching to SIM card 2, FTP connection had no response after OK was returned by **AT+QIOPEN**.
MQTT	Fixed the problem that if the message ID exceeds 65535 during MQTT message publishing, the MQTT package has been re-transmitted continuously.
MQTT	Fixed the problem that 20 K MQTT payload data could not be received normally.
NETWORK	Optimized AT commands **AT+CPOL** and **AT+CPLS**.
PSM	Optimized the URC output sequence in PSM.
TCP SSL	Fixed the problem that the configuration of **AT+QICFG="recv/ignore"** did not take effect.
TCP SSL	Fixed the problem of failed to send big data attachments through SMTP after TCP sending window size was set to 100 by **AT+QCFG="tcp/windowsize"**.
TCP SSL	Fixed the problem of incorrect URC format reported when SSL read null data.
TCP/UDP	Extended **AT+QIOPEN** to support <tcp_client_maxnum> to configure the maximum number of connections.
TCP/UDP	Fixed the problem that the module could not work normally when **ATO** command execution after using **+++** to exit TCP transparent transmission.
TCP/UDP	Optimized the reception and reporting of **AT+QISEND** data.
TCP/UDP	Solved the problem that there may be no responses after **AT+QISEND** was continuously and quickly executed.
TTS	Solved the problem of incomplete TTS resource file caused by suddenly powering down or reset during TTS resource file integration.
GENERAL	Optimized the startup logic.
GENERAL	Optimized PDP context activation.

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GENERAL	Solved the problem of repeated memory in the reset function.
GENERAL	Optimized **AT+CUSD** command.

EC200UCNAAR03A03M08	
Item	**Brief Description**
PSM	Optimized the URC output sequence in PSM.

EC200UCNAAR03A02M08	
Item	**Brief Description**
MQTT	Fixed the problem that the cache message order returned by **AT+QMTRECV?** was incorrect when MQTT message receiving was in cache mode.
MQTT	Fixed the problem that 20 K MQTT payload data could not be received normally.
UAC	Reduced UAC sampling rate from 16 K to 8 K.
GENERAL	Solved the problem that the response result from the active command PROVIDE LOCAL INFORMATION was incorrect in some specific scenarios.

EC200UCNAAR03A01M08	
Item	**Brief Description**
/	/

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3.4. Known Issues

Item	**Bug Description**
/	/

NOTE

Verification Environment is shown below. For more details, please contact Quectel technical support.
For Windows: Win10 64 bit
USB Driver: Quectel_Windows_USB_Driver(U)_V1.0.20
Qflash Tool: QFlash_V7.2

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4. Function List

Category	Item	Supported Version (Since)	Note
Basic Function	SMS	EC200UCNAAR03A01M08	/
	Voice Call	EC200UCNAAR03A01M08	/
	CMUX	EC200UCNAAR03A01M08	/
	FILE	EC200UCNAAR03A01M08	Supports UFS FILE only.
Protocol Function	TCP/ UDP	EC200UCNAAR03A01M08	/
	PPP	EC200UCNAAR03A01M08	/
	NITZ	EC200UCNAAR03A01M08	/
	FTP	EC200UCNAAR03A01M08	/
	HTTP	EC200UCNAAR03A01M08	/
	TCP SSL	EC200UCNAAR03A01M08	/
	MQTT	EC200UCNAAR03A01M08	/
	NTP	EC200UCNAAR03A01M08	/
	PING	EC200UCNAAR03A07M08	/
Special Function	SIM Detection	EC200UCNAAR03A01M08	/
	DFOTA	EC200UCNAAR03A01M08	/
	ECM	EC200UCNAAR03A01M08	/
	RNDIS	EC200UCNAAR03A01M08	/
	LowPower	EC200UCNAAR03A01M08	/
	GNSS	EC200UCNAAR03A01M08	/
	SD	EC200UCNAAR03A01M08	/
	TTS	EC200UCNAAR03A01M08	/
	Jamming Detection	EC200UCNAAR03A01M08	/

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	PSM	EC200UCNAAR03A01M08	/
	DTMF	EC200UCNAAR03A04M08	/
	DSSS	EC200UCNAAR03A05M08	/
	BT	EC200UCNAAR03A08M08	/

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With regional offices and support across the globe, our international leadership is devoted to advancing IoT and building a smarter world.

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